

UNNORMALIZED Table: OrderSystem

OrderID	OrderDate	CustomerID	CustomerName	CustomerPhone	[ProductID, ProductName, Quantity, PricePerUnit]	SalesRepID	SalesRepName	SalesRepPhone
O101	10-06-2025	C001	Areeba Khan	03001234567	[P1, Pen, 5, 10], [P2, Diary, 2, 50]	S01	Asad Raza	03211234567
O102	10-06-2025	C002	Hamna Here	03007654321	[P2, Diary, 3, 50], [P3, Marker, 1, 40]	S02	Zara Jamil	03219876543

Table: OrderDetails (1NF Form)

OrderID	OrderDate	CustomerID	CustomerName	CustomerPhone	ProductID	ProductName	Quantity	PricePerUnit	SalesRepID	SalesRepName	SalesRepPhone
O101	10-06-2025	C001	Areeba Khan	03001234567	P1	Pen	5	10	S01	Asad Raza	03211234567
O101	10-06-2025	C001	Areeba Khan	03001234567	P2	Diary	2	50	S01	Asad Raza	03211234567
O102	10-06-2025	C002	Hamna Here	03007654321	P2	Diary	3	50	S02	Zara Jamil	03219876543
O102	10-06-2025	C002	Hamna Here	03007654321	P3	Marker	1	40	S02	Zara Jamil	03219876543

STEP 2: PRIMARY KEY

Each row = one product sold in one order.

So, the **composite primary key** is:

OrderID + ProductID

STEP 3: FUNCTIONAL DEPENDENCIES (FDs)

- 1. OrderID → OrderDate, CustomerID, SalesRepID
- 2. CustomerID → CustomerName, CustomerPhone
- 3. SalesRepID → SalesRepName, SalesRepPhone
- 4. ProductID → ProductName, PricePerUnit

These are important because they include:

- Partial dependencies (violates 2NF)
- Transitive dependencies (violates 3NF)

STEP 4: SECOND NORMAL FORM (2NF)

Remove **partial dependencies** — those that depend on **only one part** of the composite key (OrderID, ProductID).

Table 1: OrderDetails (pure junction/fact table)

OrderID	ProductID	Quantity
O101	P1	5
O101	P2	2
O102	P2	3
O102	P3	1

Table 2: Order (based on OrderID)

OrderID	OrderDate	CustomerID	SalesRepID
O101	10-06-2025	C001	S01
O102	10-06-2025	C002	S02

Table 3: Customer (based on CustomerID)

CustomerID	CustomerName	CustomerPhone
C001	Areeba Khan	03001234567
C002	Hamna Here	03007654321

Table 4: Product (based on ProductID)

ProductID	ProductName	PricePerUnit
P1	Pen	10
P2	Diary	50
P3	Marker	40

Table 5: SalesRep (based on SalesRepID)

SalesRepID	SalesRepName	SalesRepPhone
S01	Asad Raza	03211234567
S02	Zara Jamil	03219876543

All **partial dependencies are removed** — this is now in 2NF

STEP 5: THIRD NORMAL FORM (3NF)

Now remove **transitive dependencies** — attributes depending on **non-key** fields.

Let's check:

- In Order table, we already separated CustomerID and SalesRepID to their own tables
- In Product, we have no transitive dependency
- In SalesRep, we removed all personal data

Already satisfied — all transitive dependencies handled in 2NF decomposition

FINAL 3NF TABLES

Table Name	Attributes
OrderDetails	OrderID, ProductID, Quantity
Order	OrderID, OrderDate, CustomerID, SalesRepID
Customer	CustomerID, CustomerName, CustomerPhone
Product	ProductID, ProductName, PricePerUnit
SalesRep	SalesRepID, SalesRepName, SalesRepPhone